TECHNICAL STRATEGY PAPER

MINING PRODUCTIVITY

Infrastructure, drilling yield and marginal-gain optimisation

A mine-to-product framework for extracting more useful material from existing assets - without confusing more drilled metres with more recovered metal.

Prepared: 2 November 2026

Scope: surface and underground hard-rock operations

Approach: operational engineering, constraint analysis and illustrative economics

Status: strategic technical framework - site validation required

CORE THESIS

Increase the conversion of each drilled metre into safely liberated ore, delivered plant feed and recovered product.

Executive summary

Mining output is a system result. A faster drill fleet can create no additional saleable output if the mine is constrained by blasting readiness, loading, haulage, a crusher, ventilation, ore handling or the processing plant. The highest-return programme therefore begins with a constraint diagnosis and then directs capital to the stage where an incremental unit of capacity has the highest marginal value.

The governing principle

Optimise tonnes of payable product per unit of total system cost - not metres drilled, truck hours or tonnes moved in isolation.

- Work backwards from recovered metal. Reconcile mined tonnes, grade, dilution, ore recovery, plant feed, metallurgical recovery and product payability.
- Create a common time and loss model. Record scheduled time, availability, utilisation, operating delays, productive rate and quality consistently across drills, loaders, trucks and fixed plant.
- Prioritise small, repeatable changes before major fleet expansion: drill accuracy, shift handover, road condition, dispatch, crusher feed discipline, planned maintenance and stockpile control.
- Use mine-to-mill blast design. Fragmentation should be optimised for shovel diggability, crusher acceptance and grinding energy together, not for a single ideal particle size.
- Build infrastructure around the bottleneck. Surface mines need road, water, power, dewatering, ROM and conveyor capacity; underground mines need access development, ventilation, pumping, ore passes, loading pockets, hoisting and communications sized as one system.
- Expect the constraint to move. A haulage improvement may expose a crusher limit; a crusher upgrade may expose mill capacity; a plant upgrade may expose ore supply or loading capacity.

What the evidence suggests

A 2025 study of ten production blasts at a Pilbara iron-ore mine reported that variable drill-pattern designs improved dig efficiency by approximately 2% with about 11% fewer holes for the dig-efficiency objective. For a fragmentation objective, the model improved adherence to target mean size by about 9% and reduced the standard deviation by about 5%, using about 13% fewer holes. The result is a site-specific optimisation case, not a transferable guarantee. [1]

Australian Government energy guidance identifies comminution as a major opportunity for energy efficiency and recommends energy-mass balances, geometallurgical characterisation, sorting, and more efficient crushing and grinding. [4] A 2026 mine-to-mill modelling paper uses more than three million simulated scenarios to connect upstream blast design choices with downstream screening, crushing, stockpiling and grinding costs. [7]

Recommended management target

Adopt one top-level measure - incremental recovered value per operating hour - supported by a balanced set of physical and financial measures. Use tonnes per hour and metal recovery as the operating truth, with cost per tonne and cost per recovered metal unit as the economic truth.

1. Measure the whole production system

The operational chain begins before the drill arrives and ends after the plant has recovered and produced a saleable concentrate, cathode, doré or other product. Every interface can destroy value: inaccurate holes, poor fragmentation, inaccessible faces, non-compliant payloads, queueing, crusher starvation, unstable feed, dilution or metallurgical losses.

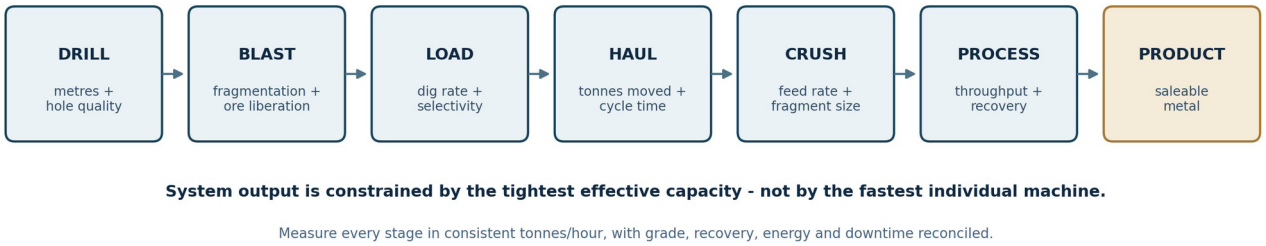


Figure 1. The mine-to-product chain. A gain at one stage only increases sustained output when downstream capacity, buffers and operating windows can absorb it.

A practical capacity model

For a first-pass diagnosis, translate every major stage into a comparable sustainable ore-equivalent capacity over the same time window:

System throughput approximation

Q_system = min(C_drill-to-blast, C_loading, C_haulage, C_crushing, C_processing, C_services)
The expression is a simplified constraint model. It assumes compatible product definitions and does not replace dynamic simulation of stockpiles, shifts, maintenance windows or changing ore blends.

The “drill-to-blast” capacity is not simply metres per hour. It is the volume or tonnage that can be reliably drilled, charged, blasted, cleared and released to production within the planning period. For a regular bench pattern, a rough geometric estimate of rock volume influenced per hole can be formed from burden × spacing × bench height; divide by drilled length and adjust for subdrill, density, pattern boundaries, geological constraints and actual recovery. Use the site’s blast design and reconciliation model for operational decisions.

From physical tonnes to recovered metal

Metal output identity

Recovered metal = mill feed tonnes × head grade × metallurgical recovery
Payable product then reflects product specification, treatment terms, impurities and commercial payability.

The distinction matters. Moving more low-grade waste is not an efficiency gain. Increasing ore tonnes while increasing dilution can lower head grade; increasing throughput can also lower recovery if residence time, grind size or reagent balance deteriorates. The correct denominator is useful, recovered material and ultimately the economic product.

Minimum data set

Stage	Measure	Why it matters
Drilling	productive metres/hour; metres per shift; hole position/depth compliance; bit life	Separates speed from effective, blast-ready metres.
Blast	fragment size distribution; oversize and fines; diggability; blast clearance time	Connects drilling quality to loading and comminution.
Loading	dig rate; bucket fill; queue / spotting time; selective mining compliance	Shows whether fragmented ore becomes clean tonnes.
Haulage	payload; loaded/empty cycle; queue; road delay; tonnes per truck-hour	Exposes distance, traffic and dispatch losses.
Plant	crusher and mill feed; availability; energy/t; recovery; product quality	Shows whether extra ROM becomes extra recovered metal.
System	ore loss; dilution; grade reconciliation; cost per recovered unit	Measures economic conversion rather than activity.

2. Marginal-gain theory: improve the current constraint

The theory of constraints gives a practical rule: find the stage that sets sustainable output, improve it, then identify the new constraint. In a mining operation, the constraint may be a physical asset, a service system, an operating rule or a geological condition. It can vary by shift, pit phase, ore type and maintenance cycle.

Worked example - illustrative open-pit mine

Assume the following sustainable stage capacities, in million ore tonnes per month. The numbers are invented for explanation and are not industry benchmarks or a forecast for any mine.

Stage capacity	Baseline	Bottleneck status
Drill / blast-ready supply	2.25	Excess capacity
Blast release	2.20	Excess capacity
Loading	2.10	Near-constraint
Haulage	1.95	Current constraint
Primary crushing	2.00	Next constraint
Processing plant	2.05	Next constraint

At baseline, the mine can sustain approximately 1.95 Mt/month because haulage is the tightest stage. Raising drill capacity by 10% does not increase the system rate: the haulage constraint remains. Raising haulage by 10% lifts available haul capacity to 2.145 Mt/month, but crushing is then the constraint at 2.00 Mt/month. Increasing crusher capacity by 10% moves the constraint to the plant at 2.05 Mt/month. Increasing plant capacity by 10% moves the constraint to loading at 2.10 Mt/month.

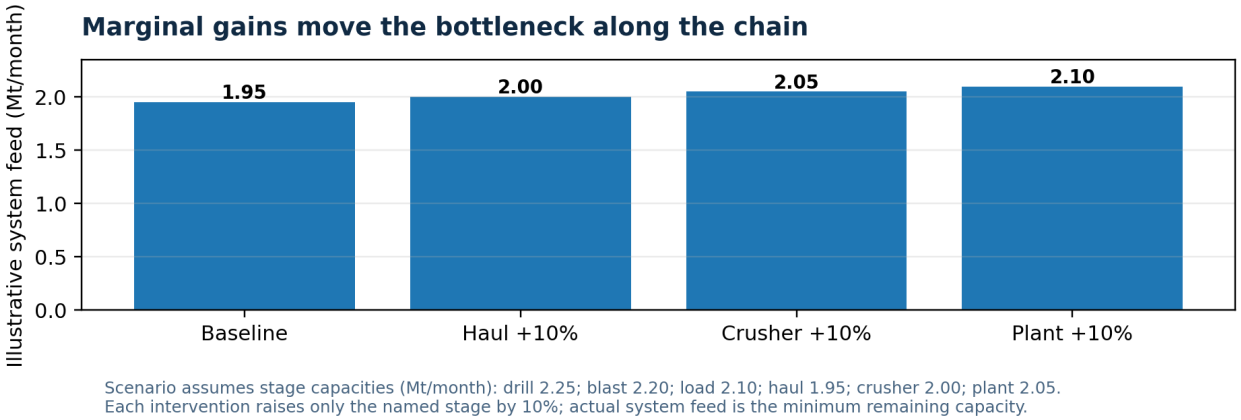


Figure 2. In this simplified sequence, the first three well-targeted interventions move throughput from 1.95 to 2.05 Mt/month (+5.1%). A drill-only expansion adds zero system throughput until a downstream constraint is addressed.

Marginal value test

For each proposed intervention, estimate:

Incremental value test

Marginal contribution = (incremental recovered product × net realised margin) - incremental operating cost - annualised capital and sustaining cost.
Then test the result against safety, environmental, geotechnical, permitting and execution constraints.

A useful screening metric is incremental recovered metal per additional operating hour and per unit of capital. For a constrained stage, estimate the derivative of system output with respect to stage capacity: a 1% increase in a non-constraining stage may have almost zero immediate value, while a 1% increase in the bottleneck can have a first-order effect until the next constraint is reached.

Why the gains do not simply add

Compound carefully

Availability × utilisation × rate × yield may compound when independent. Avoid double-counting overlapping delays; a dispatch and handover project may target the same lost hours. Pilot the change and cap gains at downstream capacity.

3. Drilling: extract more useful rock per drilled metre

Drilling performance should be reported as both metres drilled and useful blast-ready capacity. A fleet can drill quickly yet underperform if holes are misplaced, off-angle, shallow, blocked, poorly flushed or inconsistent with the geological model. The optimisation objective is reliable fragmentation and ore release at minimum total cost per recovered tonne.

High-return operational levers

Lever	Operational action	Measure of success
Drill pattern adaption	Use geotechnical and geometallurgical domains, measurement-while-drilling (MWD) response and blast history to adjust pattern geometry where justified.	Tonnes broken per drilled metre; dig rate; crusher feed PSD.
Hole accuracy	Survey collars; control inclination, azimuth and depth; audit deviations and collaring practice.	Hole compliance; redrill rate; subdrill variance; wall damage.
Consumables	Match bit and rod selection to rock abrasivity and strength; track bit life, regrinds and failure causes.	Metres per bit; cost per effective metre; non-productive change time.
Drill readiness	Pre-shift inspections, planned servicing, spare critical components and reliable support equipment.	Physical availability; mean time to repair; lost productive hours.
Water / air / flushing	Maintain pressure, flow and filtration in accordance with the rig and ground requirements.	Penetration rate stability; hole cleanliness; blockage frequency.
Workface logistics	Prepare pads, access, lighting, drainage, power/fuel and survey control ahead of the rig.	Move / setup time; productive shift share; metres per scheduled hour.
Sequencing	Coordinate drilling, charging, blasting, clearance and post-blast release with the production schedule.	Ready-to-blast inventory; blast-to-dig lead time; face starvation.

The mine-to-mill drill design

Use a feedback loop: map geology and hardness; design the pattern; capture actual hole position and MWD data; reconcile blast fragmentation; measure shovel digging, oversize, secondary breakage, crusher throughput and mill energy; update the model by geometallurgical domain. The 2025 Pilbara study by Perincek and colleagues provides a useful example of variable hole count, hole position and per-hole energy intensity being optimised together rather than imposing one uniform pattern across heterogeneous ground. [1]

Important interpretation

The target is not the finest possible fragmentation. Extra blast energy and tighter patterns add drilling, explosive and execution cost. The right fragmentation is the distribution that minimises total cost and risk from drilling through grinding while achieving required recovery and productivity.

How to measure drilling properly

Separate calendar time, scheduled time, available time, operating time and effective drilling time. Report: physical availability; utilisation when required; effective drilling rate; hole compliance; metres per bit; and tonnes released per drilled metre. Do not use penetration rate alone as a productivity KPI. Align definitions with the GMG time-use framework so avoidable idle time is not hidden inside equipment categories. [3]

4. Blasting, grade control and ore liberation

Blasting is the interface between the drill plan and the material-handling system. Its quality should be judged by the effect on excavation, dilution, crusher performance, grinding energy, recovery, vibration, flyrock, fumes and geotechnical stability. A blast with attractive fragmentation but poor ore boundary control may destroy value.

Fragmentation and downstream cost

The mine-to-mill literature consistently describes the connection between blast fragmentation and downstream energy, loading productivity and plant throughput. A 2023 review notes that grinding is particularly energy intensive and that better blast fragmentation can shift some size-reduction work upstream, where it may be achieved at lower total system cost; it also cautions that higher explosive intensity does not automatically produce better results if design and execution are poor. [5,6]

Measure a full size distribution, not just a single average. Track P80 or another site-relevant sizing metric, oversize rate, fines fraction, shovel dig time, secondary breakage, crusher blockages, recirculating load, mill specific energy and metal recovery. Image-based fragmentation data should be calibrated against physical measurements and reconciled over time.

Ore loss, dilution and destination control

Post-blast rock movement can shift ore-waste contacts. Grade-control polygons that ignore blast movement may send ore to waste or waste to the plant. Use pre- and post-blast surveys, blast movement monitoring where warranted, high-resolution grade-control models and truck-by-truck destination control. A 2024 study at the Husab Uranium Mine reported average dilution and ore-loss reductions of 1.2% and 1.5% respectively using a blast-movement monitoring workflow; treat this as a site-specific case, not a generic expectation. [10]

Practical improvements

- Reconcile planned versus actual hole position, explosive loading records, initiation timing, vibration and post-blast fragmentation by rock domain.
- Create a feedback record from each blast to shovel dig rates, oversize handling, crusher interruptions and mill energy.
- Prioritise selective blasting and ore sorting where grade contrast, mineral texture, particle size and separation response make it technically viable.
- Match blending and stockpiling to ore hardness, grade, moisture and processing response so the plant sees a stable feed envelope.
- Control flyrock, vibration, fumes, misfires, exclusion zones and wall damage through competent blast engineering and the applicable statutory system.

A useful design rule

Optimise the whole size-reduction chain. Spending a little more on drilling or blasting can be economic when it reduces secondary breakage, truck delays, crusher interruptions or grinding energy - but only when reconciled data demonstrates a net benefit.

5. Loading, haulage and material movement

In many open pits, the greatest throughput loss is not a shortage of theoretical truck capacity but wasted time in the operating cycle: spotting, queueing, waiting for a shovel, road delays, dumping constraints, refuelling, operator changeovers and poor dispatch coordination. The system must synchronise shovel capacity, truck fleet size, route geometry and crusher demand.

Loading

- Keep faces clean, drained, safely prepared and supplied with suitable fragmentation; remove avoidable obstruction and reduce machine repositioning.
- Match shovel bucket size, truck body and material density; monitor bucket passes, payload distribution, dig time, swing angle and spotting delay.
- Use grade-control and dig-limit data in the loading workflow so higher tonnage does not mean more dilution.
- Monitor shovel utilisation and queueing as a paired system; adding a truck to a shovel that is already saturated can lengthen queues without raising output.

Haulage

- Engineer haul roads for rolling resistance, drainage, width, gradients, camber, berms, intersections and turning radii; schedule grading and dust suppression against traffic demand.
- Optimise dispatch using live truck location, payload, shovel status, crusher/ROM destination, queue length and planned road closures.
- Analyse loaded and empty cycle components separately. A short reduction in queuing or dump time can outperform an expensive increase in fleet size.
- Enforce payload discipline: underloading sacrifices capacity; overloading can increase tyre, structural, safety and maintenance costs and create downstream variability.
- Assess conveyors or in-pit crushing and conveying (IPCC) where depth, haul distance, energy, mine geometry, relocatability and production life justify the capital and reduced flexibility.

Fixed plant interface

The crusher should receive controlled, consistent feed. Maintain feed-bin capacity, chute reliability, liners, grizzlies, metal detection, dust control, oversize handling and planned wear-part changes. Use a stockpile or surge bin to decouple short-duration upstream variability from continuous plant demand, but do not let the buffer conceal a persistent capacity deficit.

Underground transport

Underground systems need capacity analysis across LHDs, ore passes, grizzlies, crushers, conveyors, loading pockets, hoists and surface handling. Ore passes can reduce repeated truck movements, but long passes introduce geotechnical, wear, flow and blockage risks. A 2024 review discusses passes above 300 m as potential efficiency infrastructure while stressing the limited public evidence and the risks of hang-ups, instability and difficult inspection. [8]

6. Infrastructure upgrades that unlock production

Infrastructure should be selected as an integrated capacity package. A new drill fleet without access roads, prepared pads, power, water, maintenance support or blast-release capacity may simply shift queues. The same is true underground when more stopes are developed without ventilation, pumping, communications, ore handling or hoisting capacity.

Surface mine infrastructure

Infrastructure	What to improve	Potential production effect
Drill platforms and access	Stable, drained pads; safe turning areas; well-maintained access and short relocation routes.	Less setup time, better collar accuracy and more effective drilling hours.
Roads and intersections	Geometry, surface, drainage, dust control, berms, traffic separation and planned maintenance.	Lower cycle time and truck wear; fewer weather-related interruptions.
Power and fuel	Reliable distribution, backup strategy, charging/fuelling layout and condition monitoring.	Higher asset availability and reduced service-related idle time.
Water and dewatering	Dewatering capacity, storage, pumping efficiency, reticulation and dust-suppression planning.	More working days, safer access and more reliable drilling/processing.
ROM and crushing	Adequate dump pocket, surge capacity, feed control, conveyors, stockpile reclaim and bypass strategy.	Reduced starvation and blockage; better plant feed continuity.
Workshops and spares	Maintenance bays, lifting systems, tooling, component exchange, tyre/track support and critical spares.	Lower mean time to repair and faster return to service.
Communications	Coverage across pit, plant, dispatch and maintenance areas; resilient networks.	More accurate dispatch, telemetry and incident response.

Underground mine infrastructure

System	Design / operating focus	Typical failure mode to prevent
Declines, shafts and development	Development metres, support, services and access sequencing matched to stope schedule.	Production blocks developed too late; congestion and access bottlenecks.
Ventilation and cooling	Primary and secondary fans, regulators, ducting, refrigeration, monitoring and demand control.	Air quantity/quality or heat limits constrain equipment and shift exposure.
Pumping and drainage	Duty/standby capacity, sumps, power resilience, sediment control and water routing.	Flooding, siltation or pump failure closes working areas.
Ore passes / grizzlies	Geotechnical design, fragmentation, moisture behaviour, monitoring and safe blockage management.	Hang-ups, wear, instability and unsafe remediation.
Conveyors / hoisting	System capacity, transfer points, loading pockets, redundancy and planned inspections.	Single-point failures interrupt the entire mining circuit.
Power and communications	Reticulation, fibre/radio coverage, tracking, remote operation and emergency systems.	Loss of control, dispatch visibility or equipment support.

The Australian Government mining energy guide highlights mine-wide opportunities in pumping, ventilation, materials movement and comminution, and recommends an energy-mass balance rather than isolated equipment upgrades. [4]

7. Processing, energy and water

The mine should optimise tonnes of recovered metal per unit of energy and water, not maximise ROM feed at any cost. Processing capacity depends on ore hardness, size distribution, mineralogy, grind size, liberation, water quality, reagent regime, equipment condition and product specification.

Comminution and sorting

- Build an energy-mass balance by ore domain, distinguishing crushing, grinding, conveying, pumping and other loads. Identify fixed losses, idle running and startup/shutdown penalties.
- Use geometallurgical characterisation to connect drill-core tests and orebody domains to expected hardness, recovery, abrasion and process response.
- Assess primary/secondary crusher configuration, screening, choke feeding, liner selection, recirculating loads, mill speed, classification and grinding-media strategy.
- Evaluate HPGR, stirred mills, pre-concentration or ore sorting through site-specific testing and whole-of-circuit economics; do not assume a technology is suitable because it is efficient elsewhere.
- Use grade and hardness-aware blending to avoid sending short-term feed surges that reduce stability or recovery.

Australian Government guidance states that comminution accounts for at least 40% of total energy use in mining and mineral processing, while stressing circuit and ore-specific assessment. The same guide discusses energy-mass balances, geometallurgy, selective blasting, sorting, dewatering and ventilation. [4]

Energy and water infrastructure

Lever	Practical actions	KPI
Electrical supply	Power-quality review; transformer and motor sizing; variable-speed drives where duty profiles justify them; power-factor and demand management.	kWh/t milled; peak MW; unplanned power trips.
Pumping	Correct pump duty point; pipe diameter and friction; leak detection; variable-speed control; staged pumping.	kWh/m3 pumped; m3/t; pump availability.
Water reuse	Thickener and tailings-water recovery; segregated clean/dirty water; storage and reticulation.	Freshwater m3/t; recycle ratio; water-related downtime.
Ventilation	Airflow on demand where safe and technically appropriate; fan and duct optimisation; heat-load management.	kWh/t; air compliance; heat exposure / downtime.
Compressed air	Leak surveys, correct pressure, receiver sizing and maintenance.	kWh per unit air; leakage rate; pressure stability.
Tailings and waste	Reliable pumping, deposition planning, water recovery and geotechnical monitoring.	Availability; water return; deposition compliance.

Plant throughput caution

A faster feed rate is only a gain when product quality and metallurgical recovery remain within the operating envelope. Report both tonnes milled and recovered metal per hour, alongside specific energy and water consumption.

8. Maintenance, digital systems and automation

Reliability is a throughput strategy. The objective is to reduce unplanned downtime and shorten repair duration without simply shifting maintenance into a more expensive or unsafe pattern. A 2025 study on shovel maintainability describes continuous operation and equipment reliability as direct contributors to mine output capacity. [9] A 2026 benchmarking paper reports average utilisation of 61-65%, availability of 77-78% and OEE of 41-44% for its studied shovel-truck fleets; these values illustrate the gap between nominal capability and effective output, not a universal standard. [11]

Reliability programme

- Rank critical assets by production consequence, redundancy, repair lead time and safety consequence; focus first on constraints and single points of failure.
- Use failure coding with practical root-cause categories: component failure, operating practice, contamination, overload, design weakness, supply delay and repeat repair.
- Protect planned maintenance windows and execute work with parts, labour, permits, lifting plans and quality checks ready before shutdown.
- Track MTBF, MTTR, planned/unplanned maintenance ratio, repeat failures, schedule compliance, backlog age and parts fill rate.
- Use vibration, oil analysis, temperature and telemetry condition monitoring where failure modes are detectable and intervention lead times are actionable.

Digital operating layer

Create a common operational data model linking the mine plan, drill guidance, MWD, blast records, fleet management, payload systems, maintenance management, crusher and plant historians, laboratory assays and product reconciliation. A control-room view should show the current constraint, the next constraint, the reason for lost output, expected duration and the person accountable for the response.

Automation - in the right order

Prioritise digital time capture, reliable communications, machine health, dispatch quality and standard work before deploying more advanced autonomy. Autonomous drilling, haulage or remote operation can improve consistency or exposure management in the right setting, but introduces integration, cybersecurity, change-management, maintenance and mixed-fleet requirements. Standardise operating processes and data quality first.

Fleet measurement and benchmarking

GMG's 2020 time-classification framework provides recommended categories and KPI definitions for surface mobile equipment. Its 2024 benchmarking initiative highlights common definitions for availability, utilisation and operational delays across drills, trucks, shovels and crushers. [2,3] Adopt one measurement dictionary, time-stamped events and clear responsibility for delay coding. Benchmark only after normalising for equipment class, duty, climate, mine geometry and operating context.

9. Investment priorities and staged implementation

A disciplined programme should begin with low-capital loss removal, then selective infrastructure, then larger capacity expansion. Each project must show which constraint it addresses, what other stage becomes limiting, and the net additional recovered product available after the change.

Priority portfolio

Priority	Typical interventions	Capital profile	When to pursue
A. Operating discipline	Shift handover, delay coding, dispatch, payload compliance, road grading, drill setup, planned maintenance, ROM feed coordination.	Low	Start immediately where baseline data shows avoidable time loss.
B. Reliability and local infrastructure	Critical spares, workshop capacity, power / water reliability, drainage, crusher feed-bin reliability, radio coverage.	Low to medium	When recurring downtime or service interruptions constrain the current bottleneck.
C. Process integration	MWD-based blast domains, fragmentation measurement, ore sorting trials, stockpile / blending logic, condition monitoring.	Medium	When data indicates losses across interfaces or costly feed variability.
D. Debottlenecking capex	Additional truck / shovel capacity, road upgrades, crusher modules, conveyor changes, pumping or ventilation upgrades.	Medium to high	When a quantified constraint has positive marginal value and downstream capacity is available.
E. Major system redesign	IPCC, shaft / hoist expansion, new decline, major ore-pass system, plant expansion, electrified fleet infrastructure.	High	After integrated engineering, geotechnical, environmental and commercial studies.

30 / 90 / 365-day execution

Horizon	Work programme	Deliverable
First 30 days	Freeze KPI definitions; build the mine-to-product mass balance; identify top five losses by lost tonnes and lost metal; verify the constraint by shift and ore type.	A baseline, constraint map, ranked loss register and accountable owners.
Days 31-90	Run controlled pilots in drill accuracy, dispatch, road maintenance, crusher feed and maintenance planning; measure before/after on matched conditions.	Verified gains, cost per incremental tonne, safety review and scale-up decision.
Months 3-12	Execute infrastructure projects with short payback and clear interfaces; introduce integrated schedule and maintenance planning; update models.	Sustained performance, revised bottleneck map and approved capital packages.
Year 2 onward	Evaluate major debottlenecking, electrification, conveyors, hoisting, new access or plant capacity.	Feasibility-backed investment sequence and capacity plan.

Project approval checklist

- What is the current constraint, and what evidence confirms it across representative shifts and ore domains?
- What capacity is added at the constraint, and what is the next constraint after the project?
- What recovered product is added after dilution, recovery and product-quality adjustments?
- What additional energy, water, labour, consumables, waste and sustaining capital are required?
- What geotechnical, safety, environmental, permitting, community and closure obligations change?
- Can the gain be measured independently, with a pilot, control group or matched operating period?

10. Illustrative financial bridge and risk controls

Illustrative copper-feed example

Take the throughput sequence from Section 2. Suppose the site is a copper operation processing 1.95 Mt/month at 0.80% copper head grade and 88% metallurgical recovery. Illustrative recovered copper is:

Baseline calculation

1.95 million t × 0.80% Cu × 88% recovery = 13,728 t recovered Cu per month (before payability and commercial deductions).

At 2.05 Mt/month, with grade and recovery unchanged, the same formula gives 14,432 t/month, or 704 additional recovered tonnes per month (+5.1%). This is a purely illustrative arithmetic bridge; it is not a forecast. In practice, the incremental tonnes may have a different grade, hardness, strip ratio, recovery response, moisture content and treatment cost. The plant may also require additional power, water, reagents, tailings capacity and maintenance.

Economics to use in a real business case

Item	Treatment
Incremental saleable output	Use recovered metal or saleable product after product specification and payability.
Revenue	Apply a defensible price deck and relevant concentrate / refining terms; separate price exposure from operational gain.
Variable cost	Include drilling, consumables, explosives, fuel / power, water, tyres, labour, reagents, grinding media and waste handling.
Fixed and sustaining cost	Include additional crews, maintenance, spares, plant wear, infrastructure renewal and sustaining capital.
Capital and schedule	Include engineering, construction, commissioning, ramp-up, contingencies, taxes, closure/security and working capital.
Risk-adjusted return	Test commodity prices, grade, recovery, equipment performance, schedule slippage, water / power constraints and geotechnical conditions.

Non-negotiable controls

- Geotechnical and slope / ground-support design must govern access, drilling, blasting, ore-pass and underground development decisions.
- Blast changes must remain within statutory approvals and competent-person procedures for explosives, exclusion zones, misfires, fumes, vibration and flyrock.
- Equipment utilisation must never be raised by defeating fatigue controls, traffic separation, inspections, maintenance windows or safe operating limits.
- Water management, tailings integrity, dust, noise, biodiversity, emissions, cultural heritage and closure requirements are design constraints, not post-project adjustments.
- Digital systems require cyber controls, access governance, resilient communications, data ownership and a manual safe-state operating procedure.

Decision rule

Do not approve capacity that cannot be safely fed, serviced, powered, watered, ventilated, maintained and processed. The best marginal gain is the one that raises sustained recovered output while lowering whole-system cost and risk.

11. KPI dashboard and management cadence

A compact dashboard should show the current constraint and the performance of every linked stage. Review daily at shift level, weekly by production system and monthly through the capital and operating-cost process.

KPI family	Core measures	Review question
Drill	effective metres; hole compliance; metres/bit; productive utilisation; blast-ready tonnes/m.	Are we creating the right holes, in the right place, on time?
Blast / grade control	fragment size distribution; oversize; fines; secondary breakage; blast movement; ore loss and dilution.	Are we releasing recoverable ore with controllable fragmentation and boundaries?
Loading / haulage	dig rate; truck payload; cycle time; queue time; tonnes/truck-hour; road delay.	Can the fleet move the required tonnes without excessive idle or overload?
Fixed plant	crusher feed; crusher and mill availability; throughput; recirculating load; recovery; product quality.	Can the plant accept the material and convert it to product efficiently?
Services	power trips; water availability; pumping; ventilation compliance; communications coverage.	Are enabling systems constraining production or safe access?
Reliability	availability; utilisation; MTBF; MTTR; planned maintenance; repeat failures; backlog.	Are assets ready when required, and are repairs completed well?
Economics / ESG	cost/t ore; cost/recovered unit; kWh/t; water m3/t; waste and emissions intensity.	Did productivity create genuine net value within environmental limits?

Management cadence

Daily: review previous-shift loss events, current constraint, plant feed, ready inventory, equipment status, safety and the 24-hour plan. Weekly: reconcile actual production against the mine plan, rank recurring losses, inspect bottleneck movement and approve corrective actions. Monthly: reconcile grade, ore loss, dilution, recovery, unit cost, energy and water; update the marginal-value model and capital priorities. Quarterly: benchmark using consistent time definitions, review geometallurgical model performance and reset the next constraint programme.

Recommended north-star indicators

North-star measure	Definition
Recovered metal per operating hour	Recovered product quantity divided by actual operating hours, with grade and recovery reconciled.
Value per constrained hour	Incremental net contribution generated by one additional hour at the current system constraint.
Total cost per recovered unit	All-in operating and sustaining cost divided by recovered saleable product, using a consistent accounting boundary.
Constraint conversion efficiency	Actual system output divided by the verified capacity of the current bottleneck, adjusted for planned operating time.

Conclusions

The strongest near-term opportunity in many existing mines is not simply to drill faster or buy more trucks. It is to remove the small losses that accumulate between the mine plan and recovered product: poor hole compliance, fragmented operating data, delay at shift changes, weak road conditions, avoidable queues, crusher starvation, inconsistent feed, repeated failures and ore-boundary errors.

A marginal-gain strategy can deliver substantial cumulative improvement when each intervention is real, independent and applied to the current constraint. But gains must be measured at the system level. Increasing capacity in a non-constraining asset may create no immediate output, and every successful debottlenecking project tends to reveal the next limit.

The recommended sequence is therefore: establish a reliable mine-to-product baseline; identify and verify the constraint; improve operating discipline and asset reliability; optimise drilling and blasting with downstream feedback; strengthen the infrastructure that enables material flow; then invest in larger capacity where incremental recovered product justifies the total cost and risk.

Final proposition

More drilled metres are an input. More recovered, payable material at lower whole-system cost is the result.

Limitations

This is a cross-commodity strategic framework, not an engineering design, reserve statement, operating plan or feasibility study. The numerical example is illustrative. Equipment performance, production rates, energy intensity, recovery, dilution, capital cost and safety controls must be determined from site data, competent engineering, metallurgical testwork, geotechnical assessment and applicable law. The cited research contains study-specific findings and should not be treated as guaranteed performance.

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